

Sangwoo Seo

POSTDOCTORAL RESEARCHER @ LG AI RESEARCH.

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Education

KAIST (Korea Advanced Institute of Science and Technology)

PH.D IN INDUSTRIAL & SYSTEMS ENGINEERING

- Advisor: Prof. Chanyoung Park
- Research Interest: Graph Neural Networks

Daejeon, South Korea

Aug. 2022 - Aug. 2026

KAIST (Korea Advanced Institute of Science and Technology)

M.S IN ELECTRICAL ENGINEERING

- Advisor: Prof. Changick Kim
- Research Interest: Graph Neural Networks

Daejeon, South Korea

Aug. 2020 - Aug. 2022

DGIST (Daegu Gyeongbuk Institute of Science and Technology)

B.S. IN SCHOOL OF UNDERGRADUATE STUDIES

- Major in Electrical Engineering

Daegu, South Korea

Mar. 2014 - Aug. 2020

Position

Materials Intelligence Lab, LG AI Research

POSTDOCTORAL RESEARCHER

- Research area: Development of a Materials Foundation Model for Integrating Organic and Inorganic Materials

Seoul, South Korea

Aug. 2026 - Present

Graph and Geometric Learning Lab, Yale University

VISITING RESEARCHER

- Advisor: Rex Ying
- Research area: Circuit graph foundation model based on multimodal

New Haven, CT, USA

Nov. 2025 - Apr. 2026

Research Interests

Graph Neural Networks

- Explainability in GNN
- Multimodal learning in GNN

AI for Science, AI for engineering

- Physics AI
- Mesh Graph Simulation
- Circuit GNN

Publication

• ELASTIC: Trajectory-Based Synchronization of Event and Tracking Data in Soccer

2026

HYUNSUNG KIM, HOYOUNG CHOI, KUNHEE LEE, SANGWOO SEO, TOM BOOMSTRA, JINSUNG YOON, CHANYOUNG PARK

ACM International Conference on Information and Knowledge Management (CIKM)

• Bridging Circuit Graphs and Language: A Cross-Modal Transformer for Netlist-to-Text Alignment and Generation

2026

SANGWOO SEO, HYUNSUNG KIM, HYUNCHUL KIM, BYEONGGON KANG, REX YING, CHANYOUNG PARK

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Graph Machine Learning for the Large Model Era

• Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations

2026

SANGWOO SEO, HYUNSUNG KIM, JIWAN KIM, CHANYOUNG PARK

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Reliable Scientific Foundation Models

- **PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories** 2026
HYUNSUNG KIM, KUNHEE LEE, SANGWOO SEO, SANGKI KO, JINSUNG YOON, CHANYOUNG PARK
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
- **CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs** 2026
JIWAN KIM, KIBUM KIM, SANGWOO SEO, CHANYOUNG PARK
International Conference on Learning Representations (ICLR)
- **Better Prevent than Tackle: Valuing Defense in Soccer Based on Graph Neural Networks** 2026
HYUNSUNG KIM, SANGWOO SEO, HOYOUNG CHOI, TOM BOOMSTRA, JINSUNG YOON, CHANYOUNG PARK
MIT Sloan Sports Analytics Conference (SSAC)
- **Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction** 2025
SANGWOO SEO, JIMIN SEO, YOONHO LEE, DONGHYEON KIM, HYEJIN SHIN, BANGHYUN SUNG, CHANYOUNG PARK
International Conference on Computer-Aided Design (ICCAD)
- **Disentangling Hyperedges through the lens of Category Theory** 2025
YOONHO LEE, JUNSEOK LEE, SANGWOO SEO, SUNGWON KIM, YEONGMIN KIM, CHANYOUNG PARK
Conference on Neural Information Processing Systems (NeurIPS)
- **ELASTIC: Event-Tracking Data Synchronization in Soccer Without Annotated Event Locations** 2025
HYUNSUNG KIM, HOYOUNG CHOI, SANGWOO SEO, TOM BOOMSTRA, JINSUNG YOON, CHANYOUNG PARK
European Conference on Machine Learning and Principles and Practice Knowledge Discovery in Databases (ECML-PKDD) Workshop on MLSA
- **Global Context-aware Representation Learning for Spatially Resolved Transcriptomics** 2025
YUNHAK OH, JUNSEOK LEE, YEONGMIN KIM, SANGWOO SEO, NAMKYEONG LEE, CHANYOUNG PARK
International Conference on Machine Learning (ICML)
- **Capturing functional context of genetic pathways through hyperedge disentanglement** 2025
YOONHO LEE, JUNSEOK LEE, SANGWOO SEO, SUNGWON KIM, YEONGMIN KIM, CHANYOUNG PARK
International Conference on Learning Representations (ICLR) Workshop on Machine Learning for Genomics Explorations
- **Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck** 2024
SANGWOO SEO, SUNGWON KIM, JIHYEONG JUNG, YOONHO LEE, AND CHANYOUNG PARK
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
- **Interpretable Graph Model with Prototype-Based Graph Information Bottleneck** 2024
SANGWOO SEO, SUNGWON KIM, AND CHANYOUNG PARK
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) Workshop on Human-Interpretable AI
- **Unsupervised Episode Generation for Graph Meta-learning** 2024
JIHYEONG JUNG, SANGWOO SEO, SUNGWON KIM, AND CHANYOUNG PARK
International Conference on Machine Learning (ICML)
- **Interpretable Prototype-based Graph Information Bottleneck** 2023
SANGWOO SEO, SUNGWON KIM, AND CHANYOUNG PARK
Conference on Neural Information Processing Systems (NeurIPS) [Gold Prize at the 30th Samsung Humantech Paper Award]
- **Explanation-based Graph Neural Networks for Graph Classification** 2022
SANGWOO SEO, SEUNGJUN JUNG, AND CHANGICK KIM
International Conference on Pattern Recognition (ICPR)

Fellowships and Awards

2024	Best Paper Award at KDD 2024 Workshop on Human-Interpretable AI , ACM KDD	Barcelona, Spain
2024	KDD 2024 Student Travel Award , ACM KDD	Barcelona, Spain
2023	Gold Prize at the 30th Samsung Humantech Paper Award [KRW 20M], Samsung Electronics	Suwon, S. Korea

Reviewer Service

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**): 2024, 2026
- Conference on Neural Information Processing Systems (**NeurIPS**): 2024-2026
- International Conference on Learning Representations (**ICLR**): 2025
- Association for the Advancement of Artificial Intelligence (**AAAI**): 2026
- International Conference on Artificial Intelligence and Statistics (**AISTATS**): 2025-2026
- International Conference on Machine Learning (**ICML**): 2025-2026
- Conference on Information and Knowledge Management (**CIKM**): 2025-2026
- Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**): 2025

Experience

Graph and Geometric Learning Lab, Yale University

PROJECT PARTICIPANT

- Advisor: Rex Ying
- RAG system for financial reports

New Haven, CT, USA

Nov. 2025 - Apr. 2026

Data Science & Artificial Intelligence Lab, KAIST

PROJECT PARTICIPANT

- Partner: SK Hynix
- Development of a circuit recognition technique using graph neural networks

Daejeon, South Korea

Mar. 2024 - Nov. 2024

Computational Intelligence Lab, KAIST

PROJECT PARTICIPANT

- Partner: Satrec Initiative Analytics
- Method design for eliminating internal bias in graph neural networks

Daejeon, South Korea

Aug. 2020 - Aug. 2022

Information Communications & Security Lab, DGIST

RESEARCH INTERNSHIP

- Advisor: Prof. Jaemin Lee
- A theoretical modeling on mobile edge computing based on queuing theory and stochastic geometry

Daegu, South Korea

Jan. 2018 - Feb. 2018

Integrated NanoSystems Lab, DGIST

RESEARCH INTERNSHIP

- Advisor: Prof. Junghyup Lee
- Circuit design and production

Daegu, South Korea

Jun. 2017 - Aug. 2017

Invited Talk

- **Top Conference Session: Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck**

Yeosu, South Korea

KOREA SOFTWARE CONGRESS (KSC)

Dec. 2024

- **Flash Talk: Interpretable Prototype-based Graph Information Bottleneck**

Y-KAST INTERNATIONAL CONFERENCE

Jeju, South Korea

Mar. 2024

- **Top Conference Session: Interpretable Prototype-based Graph Information Bottleneck**

KOREA SOFTWARE CONGRESS (KSC)

Busan, South Korea

Dec. 2023

Teaching Experience

- **Teaching Assistant in Samsung Electronics**

Data Science Specialist Course (Jun. 2024)

- **Teaching Assistant in Samsung Electronics**

AI Specialist Course (Sep. 2023)

- **Teaching Assistant in KAIST**

DS503: Machine Learning for Data Science (Spring 2023)

- **Instructor in Cantho University**

Instruction in Programming Languages including Python and C (Summer 2015)

References

Chanyoung Park, Ph.D.

Associate Professor

KAIST

Department of Industrial and Systems Engineering

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Email: cy.park@kaist.ac.kr

Rex Ying, Ph.D.

Assistant Professor

Yale University

Department of Computer Science

Email: rex.ying@yale.edu